Data-Structures & Algorithms

with

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Lecture 37 (Short Lecture)

Introduction to Randomisation

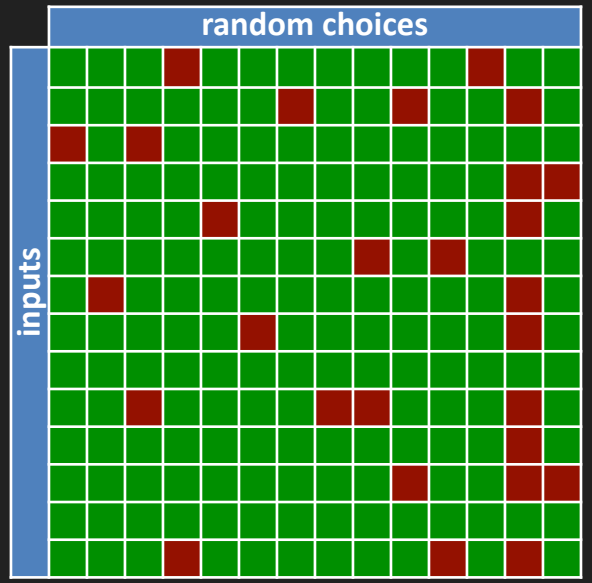
Examples already seen

Recall Two Examples

- Hash Tables
  - Recall central idea: A "universe of functions" $\mathscr{H}$ such that for any set of n keys chosen to be stored, *most* $H \in \mathscr{H}$ (say, more than 1-ε fraction) lead to low collisions
  - Without randomness, an adversary can send inputs which hash to the same bucket
- Quicksort
  - Deterministic strategies for picking pivots either result in $\Theta(n^2)$ worst-case time, or is $\Theta(n \log n)$ but with a large constant
  - Randomised pivot strategy: pick uniformly at random. $O(n \log n)$ with high probability.

Role of Randomness

- Suppose for each setting of random choice, there are bad inputs: then the worst-case performance is bad for each random choice
  - But still for each input, most of the random choices can be good!
- A deterministic algorithm obtained by fixing some random choice will be good for most inputs too
  - But it is not enough to design an algorithm to work well only when inputs are drawn uniformly or from a certain distribution



Attendance Break!

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Read the question, choose the answer and submit.
The points are mainly for attempting,
with a small bonus for getting it right.

Honour Code: Don't leave the lecture after marking attendance!

If you can't attend the whole lecture don't mark attendance!

If you are not in the classroom, a report will be sent to DADAC!

